memory needed for the tree.

9.3.4 Remove

After `search` and `insert`, we could only continue with the third basic operation on a container, `remove`. This is despite the fact that on a k-d tree, delete is not such a common operation, and some implementations don't even offer this method: as we discussed in previous section, k-d trees are not self-balancing, and so they perform best when they are created with a static set of points, avoiding frequent insertion and removal.

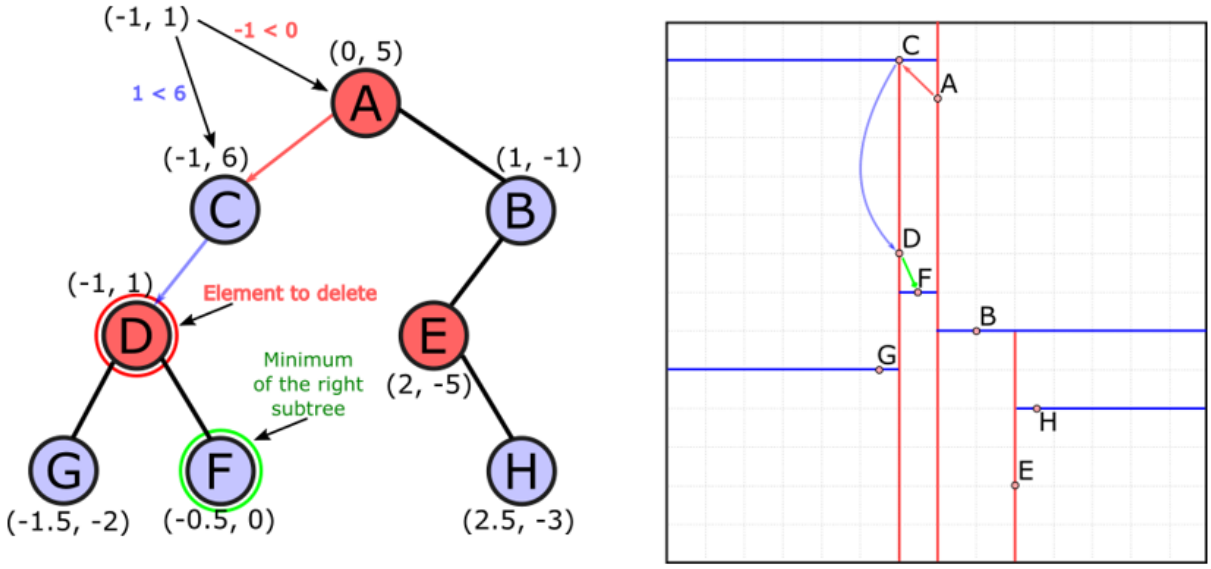


Figure 9.13 Deleting point `D` from our example k-d tree. Method `remove`, like for *BST*s, consists of a (successful) search to find the node to remove, followed, in case the node to remove is an internal node with at least one subtree, by a traversal to find an element with which it can be replaced. In this example, an internal node is removed.